

STEP 1: Business Understanding & Problem Definition

1. 1 Business Context:

This project is framed as a simulated global consulting engagement for a mid-sized omnichannel retailer operating across multiple regions. The organisation sells consumer goods through both online and physical retail channels and relies on demand forecasting to support inventory planning and order fulfilment. Recently fulfilment performance has become inconsistent, creating service issues and cost pressure.

In a competitive retail environment, service reliability directly impacts customer retention, promotion effectiveness, and margin protection, making OTIF a commercially critical metric rather than a purely operational one.

1.2 Problem Statement:

The retailer is experiencing inconsistent order fulfilment due to weak replenishment execution and supplier delivery variability, despite generally accurate demand forecasting. This has resulted in frequent stock-outs, operational inefficiencies, and reduced service reliability.

This has led to revenue leakage, increased operational firefighting, and inefficient use of working capital.

1.3 Business Objectives:

- Improve order fulfilment performance (delivery reliability and product availability).
- Reduce stock-outs and excess inventory while controlling cost.
- Improve supplier execution reliability to reduce stockouts and OTIF failures
- Improve demand forecast accuracy to support better replenishment decisions.
- Enable faster, data-driven decision-making using AI-assisted insights.

1.4 Success Metrics (KPIs):

KPI	It Measures	It Matters
MAPE	Forecast Accuracy error	Lower error improves planning
OTIF	On-time and in-full delivery	Service level & reliability
Fill Rate	% demand fulfilled from stock	Availability & customer impact
Stock-out Rate	Frequency out of stock	Lost sales and dissatisfaction
Inventory Turnover	Efficiency of inventory usage	Working capital and cost

OTIF definition used in this project: “OTIF_Flag = 1 only when delivery is on-time AND demand is fully met (no stockout).”

1.5 Scope:

In Scope: Demand Forecasting Analysis, fulfilment KPI analysis, process mapping , dashboard design, recommendations and implementation roadmap.

Out of Scope: ERP implementation, real-time system automation , supplier contract renegotiation.

1.6 Assumptions:

- Historical demand data represents typical patterns with seasonal peaks.
- Lead times vary slightly but remain within an expected range.
- The focus is on planning and decision support, not full automation.
- KPIs are measured monthly for consistency across regions.

STEP 2: Stakeholder Analysis

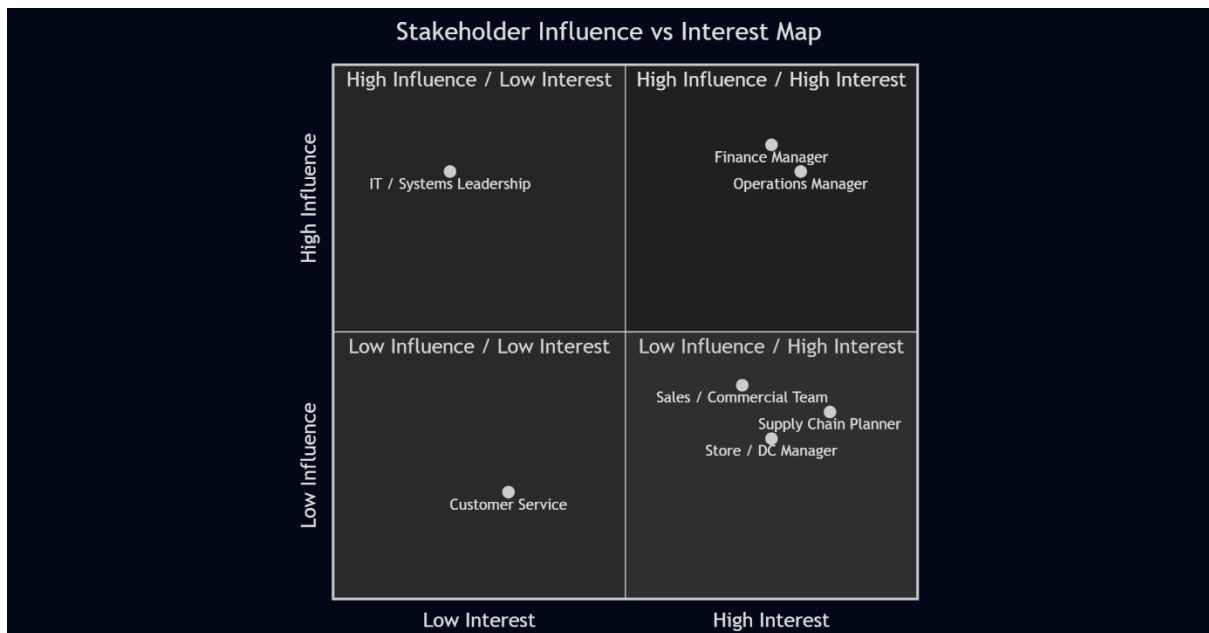
2.1 Stakeholder Overview:

As this is a simulated consulting case, stakeholders are defined as typical roles found in global retail operations and supply chain environments. The analysis focuses on how each role is impacted by forecasting accuracy and order fulfilment performance, and what information or outcomes they require to make decisions

2.2 Stakeholder Register:

Stakeholder Role	Influence	Interest	What they care about	Typical decisions
Operations Manager (Product Owner)	High	High	OTIF, service level, delivery delays	Prioritise fulfilment improvements
Supply chain Planner	Medium	High	Forecast accuracy, replenishment, stock-outs	Order quantities. Reorder timing
Store / DC Manager	Medium	Medium	Pick/Pack efficiency, availability	Staffing, Operational fixes
Finance Manager	High	Medium	Inventory cost, working capital	Budget, cost controls
Sales/ Commercial	Medium	High	Availability, lost sales	Promotions planning, demand insights
BI/ Data Analyst	Medium	Medium	Data quality, dashboard accuracy	Reporting logic, KPI definitions
IT/Systems	Medium	Low-medium	Integration feasibility, governance	Tool support, access approvals

Customer service	Low- Medium	Medium	Complaints, refunds, delivery issues	Escalations, customer feedback loop
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2.3 Stakeholder-KPI Matrix:

Stakeholder Role	Primary KPIs	KPIs matter to them
Operations Manager	OTIF, Fill rate	Overall service reliability
Supply chain Planner	MAPE, Stock-out Rate	Better replenishment decisions
Store/ DC Manager	OTIF (component) , Stock-outs	Day-to-day execution issues
Finance Manager	Inventory Turnover, Stock-outs(cost-impact)	Working capital + waste reduction
Sales / Commercial	Fill Rate, Stock-outs	Revenue protection & promotions
BI / Data Analyst	KPI consistency, data quality checks	Trust worthy reporting
Customer Service	OTIF, Stock-outs	Complaint and refund reduction

2.4 Stakeholder Needs:

- Operations Manager needs weekly visibility of OTIF and exceptions (late/ incomplete orders).
- Supply Chain Planner needs forecast vs actual trends and recommended replenishment quantities.

- Finance needs inventory value trends and proof of cost reduction opportunities.
- Sales needs visibility of availability risk during promotions and peak periods.
- A single source of truth dataset and consistent KPI logic to avoid conflicting numbers across Excel and dashboards.
- Decision authority for fulfilment changes sits with Supply Chain Leadership, supported by operational insights produced by the analytics and BI teams.
- This clarity of ownership ensures that insights generated by analytics are translated into operational decisions rather than remaining as passive reports.

STEP 3: Process Mapping – Current State

3.1 Purpose of Process Mapping :

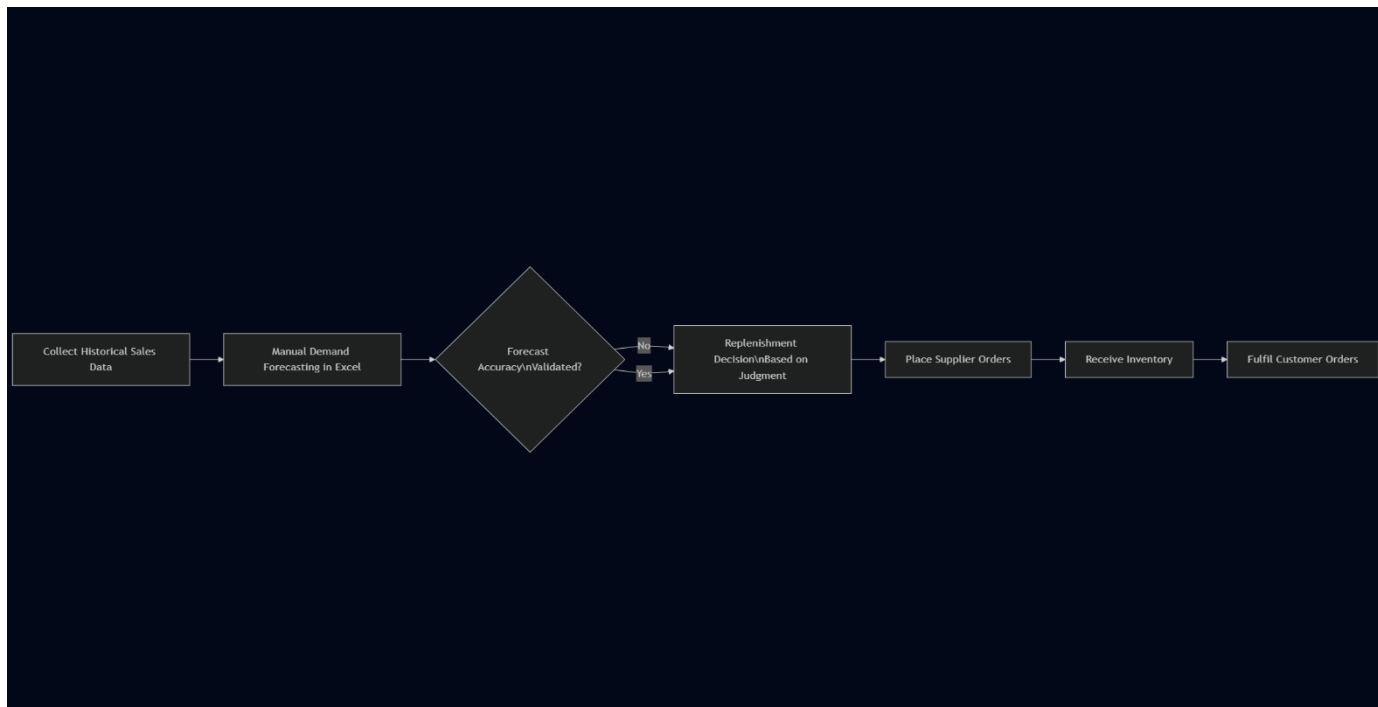
The purpose of the As-Is process mapping is to document the current order fulfilment and demand forecasting workflow used by the organisation. This provides a shared, cross-functional view of how work is currently performed and where control gaps and execution risks arise.

3.2 Process Scope:

The As-Is process begins with the collection of historical sales data and ends with customer order fulfilment. It covers forecasting, replenishment planning, inventory availability, and delivery execution.

3.3 High-Level As-Is Process Steps:

1. Historical sales data is collected from sales systems.
2. Demand forecasting is performed manually using spreadsheets.
3. Forecasts are created in spreadsheets with limited accuracy tracking and no standard governance.
4. Replenishment decisions rely on planner judgement rather than rule-based thresholds and exception reporting.
5. Orders are placed with suppliers.
6. Inventory is received and stored.
7. Customer orders are fulfilled from available inventory.
8. Performance is reviewed reactively after service failures occur, with limited root-cause learning.



3.4 Pain-Point Analysis:

Process Step	Issue Identified	Business Impact
Manual Forecasting	High Manual Effort, no consistency	Forecast Errors
Forecast Validation	No formal accuracy tracking	Poor planning Decisions
Replenishment	Judgement based decisions	Stockouts/overstock
Inventory Availability	Limited Visibility	Missed Sales
Fulfilment	Reactive issue handling	Poor Service Levels

3.5 As-Is Process Summary:

The current process relies heavily on manual forecasting and judgement-based replenishment decisions. The lack of forecast validation, limited KPI visibility, and reactive fulfilment management contribute to inconsistent service levels and inefficient inventory usage.

3.6 Controls and Risks (As-Is):

- Control gap: KPI definitions not standardised (risk of conflicting reporting).
- Control gap: No exception triage, leading to firefighting.
- Risk: Supplier under-delivery not measured consistently.

3.7 BPMN Model — As-Is Order Fulfilment Process

The current workflow was modelled using BPMN to provide a standard, cross-functional view of activities, decision points, and handoffs. The As-Is BPMN highlights manual spreadsheet forecasting, judgement-based replenishment, and reactive exception handling, which contribute to inconsistent service levels.

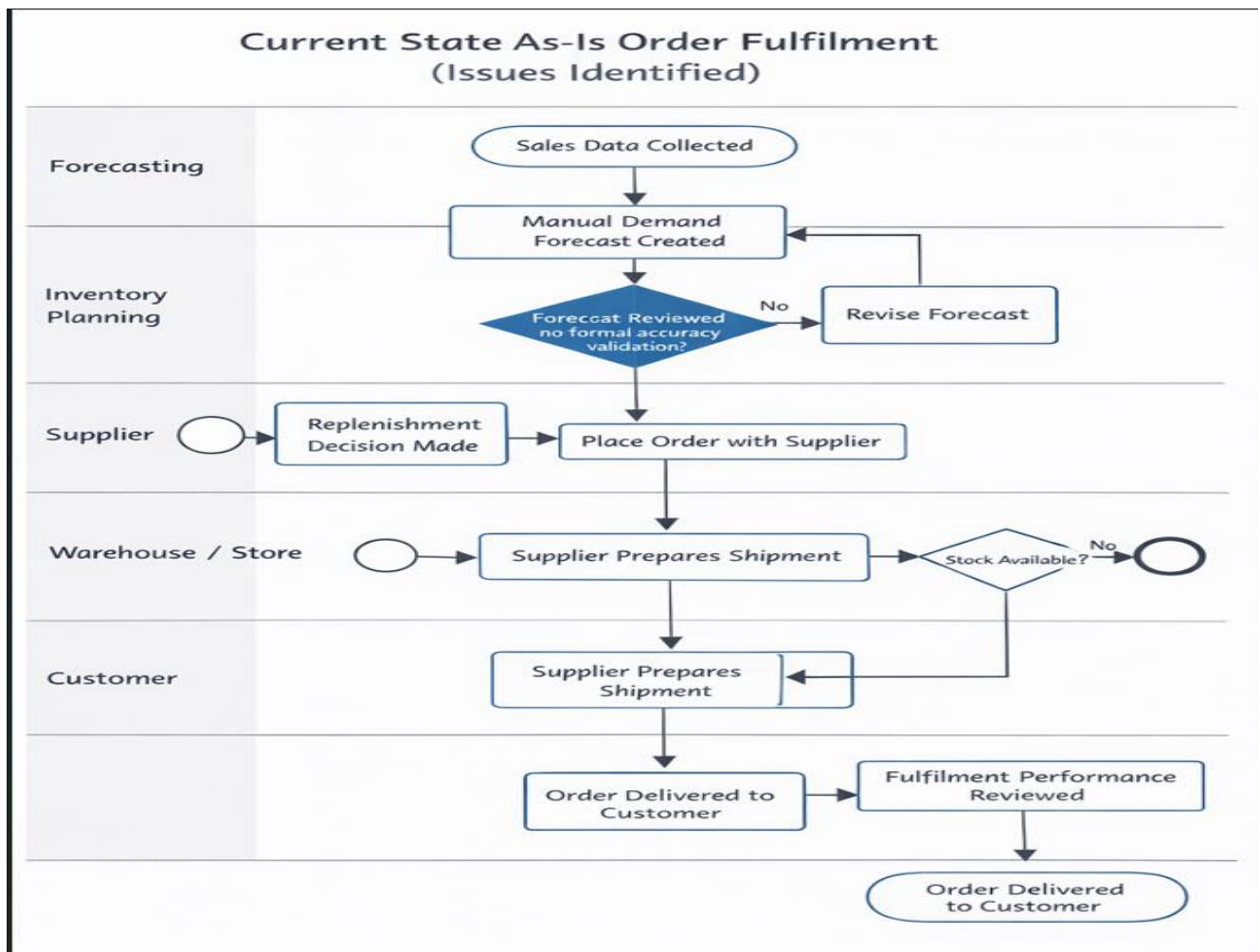


Figure 3.1 illustrates the current (“As-Is”) order fulfilment process, highlighting manual spreadsheet-based forecasting, judgement-driven replenishment decisions, limited forecast accuracy validation, and reactive exception handling. The BPMN model exposes control gaps and hand-off inefficiencies that contribute to frequent stock-outs and inconsistent OTIF performance.

Step 4: Process Mapping – Future State

4.1 Purpose of To-Be Process Mapping:

The To-Be process redesign introduces a forecast-driven planning approach, KPI visibilities and governance routines to improve order fulfilment reliability. The future state focuses on reducing manual rework, validating forecast performance, and enabling faster decisions through standard dashboards and exception-based alerts.

The design deliberately avoids full automation, focusing instead on governance, visibility, and decision quality.

4.2 Product Goals and Non-Goals:

Product goals

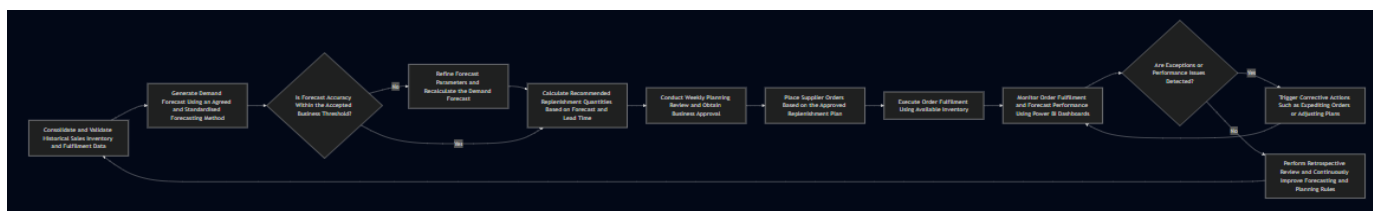
- Provide reliable forecasts with accuracy tracking (MAPE)
- Standardise replenishment decisions (rules + review cadence)
- Improve OTIF and reduce stock-outs through early visibility
- Enable stakeholders to act using dashboards and exceptions

Non-goals (for this phase)

- Full ERP replacement or end-to-end automation
- Real-time optimisation engine
- Supplier contract re-negotiation

4.3 High-Level To-Be Process Steps:

1. Consolidate and validate sales, inventory, and fulfilment data (single trusted dataset).
2. Generate forecast using a defined method (baseline + improved model).
3. Validate forecast accuracy (MAPE/MAD) and log exceptions.
4. Calculate replenishment recommendations (reorder point + safety stock assumptions).
5. Review recommendations in a weekly planning meeting (business + ops).
6. Place supplier orders based on approved plan.
7. Track fulfilment KPIs (OTIF, Fill Rate, Stock-outs) via Power BI dashboard.
8. Trigger exception actions (e.g., stock-out risk, demand spikes, delivery delays).
9. Continuous improvement: update parameters/model based on outcomes.
10. KPI definitions, ownership, and refresh cadence are formally documented and governed.



4.4 Process Improvement Summary:

Pain Point (As-Is)	To-Be Change	Expected benefit
Manual forecasting	Standard method + defined inputs	Consistency & less rework
No accuracy validation	Track MAPE/MAD, log exceptions	Better planning decisions
Judgment-based replenishment	Rules + weekly review	Fewer stock-outs/overstock
Low KPI visibility	Power BI dashboards	Faster, data-driven action

Reactive fulfilment fixes	Exception-based alerts/actions	Reduced service failures
No learning loop	Retrospectives & parameter updates	Continuous improvement

4.5 MVP vs Future Enhancements:

MVP (first release)

- One trusted dataset + basic validation checks
- Forecast model with accuracy tracking (MAPE)
- Core KPI dashboard (OTIF, Fill Rate, Stock-outs)
- Weekly review cadence + exception log

Next iterations

- Category/SKU segmentation and improved seasonality handling
- Automated alerts (email/Teams) for exceptions
- Scenario planning (promotion uplift, lead-time changes)
- Advanced replenishment optimisation (optional)

4.6 KPI Thresholds:

- OTIF target trigger: $OTIF < 95\% \rightarrow$ open exception.
- Stockout trigger: $Stockout_Flag = 1 \rightarrow$ open exception.
- On-time trigger: $OnTime < 90\% \rightarrow$ review carrier/supplier.
- Forecast trigger: $MAPE > 10\% \rightarrow$ recalibrate model.

4.7 BPMN Model — To-Be KPI-Driven Fulfilment Process

A To-Be BPMN model was designed to embed KPI governance, exception-based gateways, and dashboard-driven monitoring into operational flow. This ensures OTIF and stockout risks are identified early and resolved through structured actions rather than reactive firefighting.

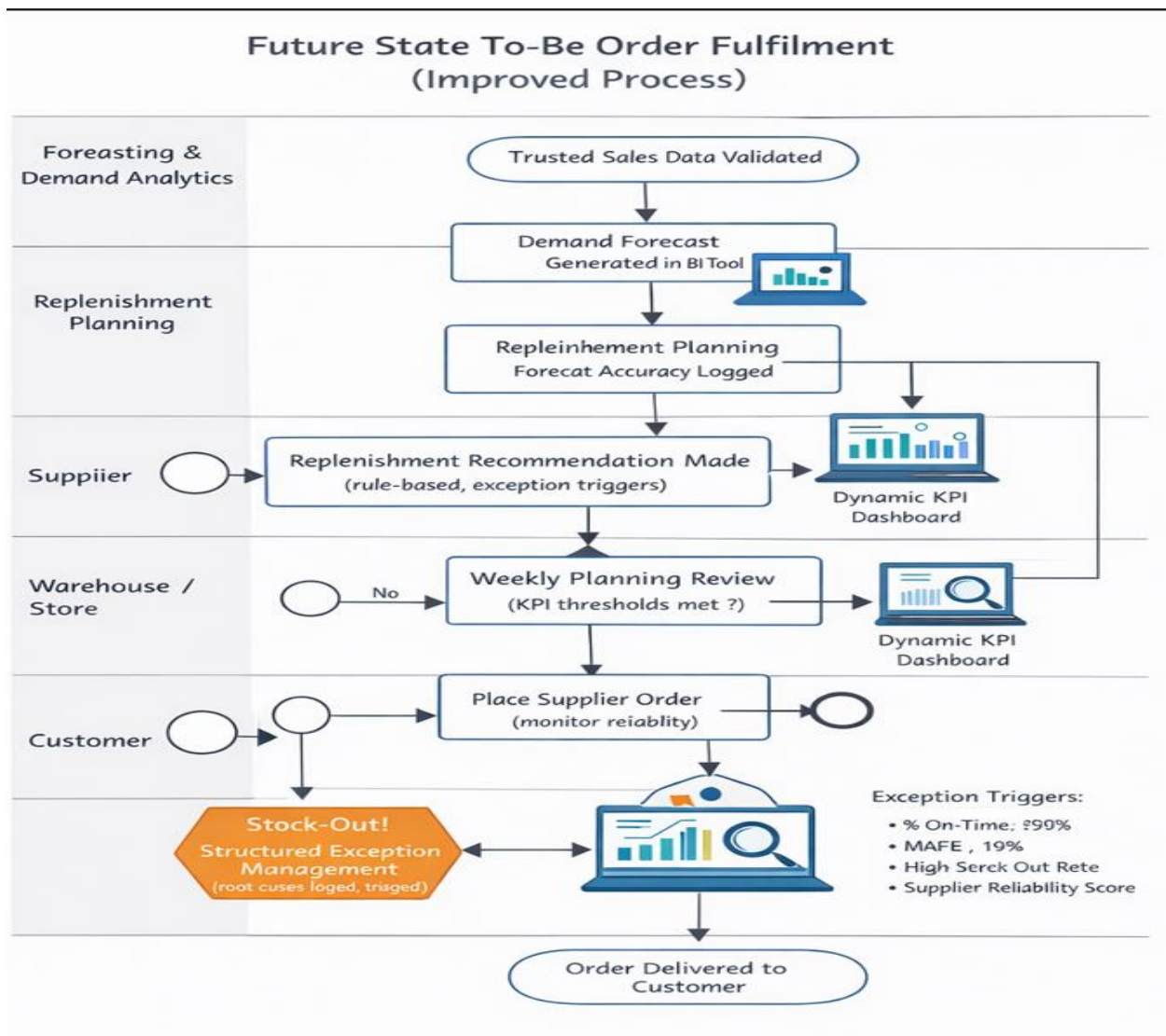


Figure 4.1 presents the redesigned (“To-Be”) order fulfilment process, embedding forecast accuracy tracking, rule-based replenishment, Power BI-driven KPI monitoring, and structured exception management. The model demonstrates how early visibility of OTIF and stock-out risks enables proactive decision-making and continuous improvement.

Step 5: Requirements and Agile Setup (Sprint 0)

5.1 Agile Delivery Approach:

This project is delivered using Agile Scrum to enable incremental value delivery and early stakeholder feedback. Work is structured into a short Sprint 0 for discovery and setup, followed by sprints 1-4 focusing on forecasting, KPI measurement, dashboard delivery, and actionable recommendations. The approach prioritises an MVP that delivers basic forecasting accuracy tracking and fulfilment KPI visibility before adding enhancements.

5.2 Scrum Roles

Product Owner: Operations Manager (Prioritises outcomes: OTIF, stock-outs)

Scrum Master : Project lead (removes blockers, ensures cadence)

Delivery Team: Business Analyst, BI analyst, data analyst

5.3 MVP Definition:

The MVP is considered successful if it delivers:

1. A clean, documented dataset with agreed KPI definitions.
2. A forecast model with measured accuracy (MAPE)
3. A basic replenishment recommendation logic
4. A Power BI dashboard showing key fulfilment trends, On-Time Delivery, monthly order volumes, revenue, and decision-support visuals for operational monitoring.
5. A weekly review cadence and an exception log

5.4 Epics:

Epic ID	Epic Name	Description	Business Value
E1	Data Foundation & Governance	Define data structure, cleaning rules, KPI definitions	Trustworthy decisions
E2	Forecasting & Accuracy Tracking	Build baseline + improved forecasts, track MAPE/MAD	Better planning
E3	Replenishment Decision Support	Translate forecast into reorder recommendations	Reduced stock-outs/overstock
E4	KPI Monitoring Dashboard	Design and deliver a Power BI dashboard for fulfilment and forecasting KPIs	Visibility & control
E5	Exception Management & Continuous Improvement	Define exception log, review cadence, improvement loop	Faster response

5.5 User Stories and Acceptance Criteria:

OTIF	As a	I want	So that	Acceptance Criteria(Given/when/Then)
US1	Supply Chain Planner	A standard dataset Template	Data is consistent	Given raw data when cleaned. Then missing

			across months	values are flagged and a clean table is produced
US2	BI Analyst	KPI Definitions documented	Dashboards are consistent	Given KPI list, when calculated, then formulas and definitions are recorded in a data dictionary
US3	Operations Manager	OTIF and Fill Rate tracking	I can monitor service	Given order data, when KPIs computed, then OTIF/Fill rate are available monthly by category
US4	Supply Chain Planner	Baseline forecast(native/moving average)	I have a benchmark	Given historical demand, when baseline methods run, then forecasts are produced for next period
US5	Supply Chain Planner	Exponential Smoothing Forecast	Accuracy improves	Given demand series, when alpha changes, then forecast updates and MAPE recalculates
US6	Operations Manager	Forecast Accuracy view	I can trust Planning	Given forecasts and actuals, when compared, then MAPE is shown over time
US7	Supply Chain Planner	A Simple replenishment Recommendation	Ordering is less reactive	Given forecast and lead time, when reorder point logic is applied, then suggested order quantity is generated
US8	Operations Manager	A Power BI Dashboard	Decisions are Faster	Given clean data, when loaded to Power BI, then KPI cards and trend visuals render correctly with filters.
US9	Operations Manager	An exception log	Issues are tracked	Given KPI threshold, when breached, then exception is recorded with date, reason, action
US10	Product Owner	Sprint demos	I can confirm values	Given sprint increment, when review happens, then working outputs are demonstrated and feedback is captured

5.6 Sprint Plan:

Sprint	Goal	Key Deliverables	Demo Output
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Sprint 0	Align on Scope & Setup	Charter, KPIs, backlog, data template	Ready to build pack
Sprint1	Baseline Forecasting	Clean dataset + baseline forecast + MAPE	Excel model v1
Sprint2	Improve forecasting + replenishment	Smoothing forecast + recommendation logic	Excel model v2
Sprint 3	Dashboard Delivery	Power BI dashboard + KPI trends	Dashboard demo
Sprint 4	Decision Support	Insights recommendations, implementation plan	Final Advisory Pack

5.7 Definition of Ready (DoR):

A story is “Ready” when: data is available, acceptance criteria written, dependencies known, estimated, and fits within sprint capacity.

5.8 Definition of Done (DoD):

A story is “Done” when: output is created, validated, documented, reviewed, and included in the sprint demo with evidence (screenshots/tables).

5.9 Risks & Dependencies:

- Dependency: clean data availability (Sprint 0)
- Risk: KPI mismatch between Excel and Power BI
- Mitigation: single KPI definition page + reconciliation checks

5.10 Requirements Traceability

Epic	User Stories	Output
E1	US1-US2	Clean dataset + KPI definitions
E2	US4-US6	Forecast model + accuracy
E3	US7	Replenishment logic
E4	US8	Power BI dashboard
E5	US9-US10	Exception log + sprint demos

Step 6: Data Understanding & Preparation.

6.1 Data Scope:

The dataset is designed to support forecasting accuracy measurement and fulfilment KPI tracking at a monthly level. It includes demand, inventory, ordering, delivery outcomes, and lead-time indicators required to evaluate OTIF, Fill Rate, Stock-outs, and forecast error(MAPE)

This project uses two analytical data layers. The first is a simulated retail operations dataset used to support forecasting, KPI modelling, and root-cause analysis. The second is an external benchmark fulfilment dataset used to analyse delivery performance at scale. These datasets are not merged and are used for different analytical purposes. In addition, a monthly aggregated summary table is used to support Power BI dashboard reporting.

6.2 Data Dictionary:

Field	Definition	Type	Example (from your data)	Notes
Month	Reporting month of operations	Date (YYYY-MM)	2024-01	Monthly aggregation
Category	Product category	Text	Grocery	Single category used for MVP
Demand_Units	Actual customer demand in units	Number	1500	Used as ground truth
Forecast_Units	Forecasted demand generated for the month	Number	1485	Close to demand, not exact
Inventory_Start_Units	Inventory available at start of month	Number	1800	Includes safety buffer
Orders_Placed_Units	Units ordered from suppliers	Number	1700	Planning decision
Orders_Delivered_Units	Units actually delivered by suppliers	Number	1525	May be less than placed
OnTime_Flag	Indicates if delivery was on time	Binary (0/1)	1	1 = On time, 0 = Late
LeadTime_Days	Supplier lead time in days	Number	9	Ranges between 7–14

Field	Definition	Type	Example (from your data)	Notes
Stockout_Flag	Indicates whether stock-out occurred	Binary (0/1)	1	1 = Delivered < Demand

6.3 Data Cleaning and Validation Rules:

- Duplicate records are removed based on the combination of **Month and Category**.
- Month values are standardised to **YYYY-MM** format.
- All numeric fields must contain **non-negative values**.
- **Any negative values are set to null and flagged as a data error**
- If Orders_Delivered_Units > Orders_Placed_Units, delivered units are capped at placed units and logged as a data exception
- If Orders_Delivered_Units < Demand_Units, Stockout_Flag is set to **1**
- Rows with missing values in critical fields (Month, Demand_Units, Forecast_Units) are flagged for review and excluded from KPI calculations

6.4 Data Quality Checklist:

The dataset is accepted for analysis only if:

- All rows contain valid values for **Month, Category, Demand_Units**
- No duplicate Month–Category combinations exist
- Demand_Units and Forecast_Units fall within a reasonable retail range (**0–5000 units**)
- OnTime_Flag and Stockout_Flag contain only **0 or 1**
- LeadTime_Days lies within the expected operational range (**1–30 days**)
- Orders_Delivered_Units do not exceed Orders_Placed_Units

6.5 Assumptions Log:

Assumption	Rationale	Impact
Monthly data aggregation is sufficient	Aligns with strategic planning cycles	Limits daily operational granularity
Single category (Grocery) used for MVP	Keeps project scope manageable	Findings may vary by category
Inventory includes a fixed safety buffer	Reflects service-level focused policy	May not reflect optimised safety stock policies

Assumption	Rationale	Impact
Delivered units may be less than placed	Simulates supplier variability	Enables realistic stock-out analysis
Dataset is simulated but designed to reflect realistic retail operating conditions.	Real retail data is confidential	Focuses assessment on method and insight

Future iterations will extend analysis to SKU and supplier level.

Step 7: Excel Analysis – Forecast Accuracy and Fulfilment KPIs

7.1 KPI Target:

KPI	Current (from Excel)	Target (PM assumption)	Gap
MAPE	3%	$\leq 10\%$	Exceeds Target
OTIF	25%	$\geq 95\%$	-70%
Fill Rate	98%	$\geq 97\%$	Meets Target
Stockout Rate	63%	$\leq 3\%$	+60%

MAPE, MAE, and Bias are calculated using forecast error logic based on the difference between Demand_Units and Forecast_Units, as documented in the analytical model and supporting evidence in Appendix C.

Note: OTIF in this project is calculated at period level using OTIF_Flag, where OTIF_Flag = 1 only when OnTime_Flag = 1 AND Stockout_Flag = 0 (on-time and in-full).

Targets represent indicative benchmarks used for analytical comparison and prioritisation rather than committed operational targets.

The contrast between strong forecast accuracy and weak fulfilment outcomes was the key analytical signal driving the remainder of the project.

7.2 KPI Interpretation:

Based on the KPI Summary derived from the Excel analysis:

- **Forecast accuracy is strong**, with a MAPE of **3%** and MAE of **48.4 units**, indicating that demand is being predicted reliably.
- **Forecast Bias of -10.1 units** suggests slight over-forecasting on average, but this is not material enough to cause major inefficiencies.

- **Fulfilment performance is weak**, with OTIF at only **25%**, despite an On-Time Delivery rate of **79%**.
- **Stockout Rate is very high (63%)**, indicating that in a majority of months, delivered quantities were insufficient to meet customer demand.
- **Fill Rate remains high at 98%**, implying that although shortages occur frequently, their magnitude is often small rather than severe.
- These KPI results form the basis of the project's operational diagnosis and are interpreted alongside benchmark fulfilment analysis rather than in isolation.

7.3 Root Cause Analysis:

Using the KPI results, the root causes can be summarised as follows:

Observation	Root Cause
Low OTIF (25%)	Combination of late deliveries and frequent stockouts
High Stockout Rate (63%)	Supplier under-delivery relative to demand
High Fill Rate (98%)	Shortfalls are usually marginal, not catastrophic
Good Forecast Accuracy	Demand planning capability is mature

7.4 Business Impact

From a business perspective, the current performance implies:

- **Customer dissatisfaction risk** due to frequent stockouts, even if quantities are small.
- **Revenue leakage**, as unmet demand cannot always be recovered.
- **Operational inefficiency**, with planners forecasting accurately but suppliers failing to execute.
- **Management blind spots**, as high fill rate may mask the true frequency of service failures.

This combination is particularly risky in a retail environment where **service reliability** drives repeat purchases.

7.5 Initial Improvement Opportunities (from KPI diagnosis):

Short-term (0–3 months) – Execution fixes

- Introduce exception triggers when Stockout_Flag = 1 or Fill Rate < 95%
- Conduct supplier performance reviews for months with late or short deliveries
- Implement weekly OTIF monitoring instead of monthly review

Medium-term (3–6 months) – Process improvements

- Introduce dynamic safety stock rules based on historical delivery reliability
- Segment suppliers by lead-time and reliability performance
- Align replenishment quantities more closely with execution risk, not just demand

Long-term (6–12 months) – Capability uplift

- Integrate AI-assisted exception forecasting to flag high-risk months
- Move from static replenishment to adaptive planning
- Embed KPIs into a Power BI operational dashboard for leadership visibility

7.6 Role Of AI :

While forecasting accuracy is already strong using traditional methods, AI-assisted analysis can be used in subsequent phases to identify exception patterns, supplier risk signals, and demand anomalies earlier. In this project, AI is positioned as a decision-support tool to enhance human judgement rather than replace planning logic.

7.7 Final Summary:

This demonstrated that while demand forecasting performance is strong, operational execution significantly limits service performance. The analysis highlights that improvements in supplier reliability, replenishment execution, and exception management will deliver greater business impact than further refinement of forecasting models alone.

Therefore, improving execution reliability will deliver higher ROI than further forecast optimisation.

STEP 8 : Define the AI Use-Case

8.1 AI Use-Case Definition:

AI is used to support business analysis by identifying exception patterns, highlighting risk signals, and generating decision-support insights. Traditional Excel-based KPI analysis remains the primary source of truth for operational measurement (MAPE, OTIF, Fill Rate). In addition, SQL-based analysis on an external e-commerce dataset (Olist) was used as a benchmarking layer to compare fulfilment patterns at scale. These SQL KPIs are not combined with the Excel KPIs; they provide external context for what 'good' delivery performance looks like.

8.2 AI Input Pack:

Item	What you provide
KPI Summary	MAPE 3%, OTIF 25%, Fill Rate 98%, Stockout 63%, On-time 79%

Item	What you provide
Exception months	Months where Stockout_Flag = 1 or OTIF_Flag = 0
Variables	Forecast_Error, LeadTime_Days, OnTime_Flag, Delivered vs Demand
Business question	“Why is OTIF low despite strong forecast accuracy?”

8.3 AI Assisted Exception Detection:

AI was used to propose likely drivers behind low OTIF performance. The model suggested that OTIF degradation is primarily linked to stockouts and late delivery incidents rather than demand forecast error. It also flagged that lead time variability and delivered-quantity shortfalls are frequent in months where OTIF_Flag = 0.

8.4 Validate AI Outputs Against Excel Data Evidence:

AI Claim	Evidence from Excel	Decision
“Forecast error is not the main OTIF driver.”	MAPE is 3% (strong accuracy), OTIF is 25% (weak).	✓ Accept
“Stockouts are frequent and reduce OTIF.”	Stockout Rate is 63%, OTIF_Flag often 0 when Stockout_Flag = 1.	✓ Accept
“Late deliveries contribute to OTIF failure.”	On-Time is 79%, OTIF (25%) is much lower → late delivery + stockouts both matter.	✓ Accept
“Lead time variability signals risk.”	LeadTime ranges 7–14; correlate with exception months (review required).	⚠ Partially accept
“AI recommends increasing inventory buffers.”	Inventory is already Demand + 300; stockouts still occur due to under-delivery.	✗ Reject (root cause is supplier delivery)

8.5 AI vs Traditional Forecasting:

Traditional forecasting performance was already strong (MAPE 3%). Therefore, AI was not used to replace the forecast model but to enhance decision-making through exception forecasting: identifying high-risk months where operational execution may fail despite accurate demand prediction.

8.6 AI-Enabled Enhancements:

AI Enhancement	Value	Priority
Exception Risk Score (0–100) using stockout + on-time + lead time	Predict OTIF failure early	High
Supplier reliability classifier	Identify supplier risk trends	High
Automated root-cause summary (monthly)	Faster ops reviews	Medium
Demand anomaly detection	Flag unusual demand spikes	Medium
What-if simulator (lead time / delivery %)	Support replenishment decisions	Medium

All AI-enabled outputs are treated as advisory signals and remain subject to human validation, ensuring transparency, accountability, and operational control.

8.7 SQL Findings & KPI Interpretation:

SQL-based fulfilment analysis shows that approximately **90.45% of customer orders are delivered on time**, indicating a largely reliable logistics and delivery operation. The OTIF-like metric mirrors this value due to the absence of inventory quantity data, suggesting that fulfilment challenges are primarily related to delivery timing rather than product availability.

The average delivery delay across all orders is 0.72 days, with late orders experiencing higher delays. This indicates that most delivery delays are relatively small in overall operational terms, even though late orders remain a service concern. SQL results are used for benchmarking only and are not merged with the operational KPI dataset. Compared to the simulated dataset (On-Time 79%, OTIF 25%), the project's improvement focus remains on execution reliability (supplier under-delivery + late deliveries), supported by exception management and decision-support analytics.

	month text	total_orders bigint	total_revenue numeric	avg_order_value numeric	on_time_percent numeric
1	2016-09	4	354.75	88.69	0.00
2	2016-10	324	56808.84	175.34	82.72
3	2016-12	1	19.62	19.62	100.00
4	2017-01	800	137188.49	171.49	91.00
5	2017-02	1780	286280.62	160.83	90.11
6	2017-03	2682	432048.59	161.09	90.60
7	2017-04	2404	412422.24	171.56	89.52
8	2017-05	3700	586190.95	158.43	92.95
9	2017-06	3245	502963.04	155.00	93.68
10	2017-07	4026	584971.62	145.30	93.49
11	2017-08	4331	668204.60	154.28	94.00
12	2017-09	4285	720398.91	168.12	92.60

The following screenshot presents the SQL-based computation of fulfilment KPIs derived from the order_analytics view, including On-Time Delivery percentage, OTIF-like performance, and average delivery delay.

```
SELECT
  COUNT(*) AS total_orders,
  ROUND(AVG(on_time_flag::numeric) * 100, 2) AS on_time_percent,
  ROUND(
    AVG(
      CASE
        WHEN delivered_date IS NOT NULL AND on_time_flag = 1 THEN 1
        ELSE 0
      END::numeric
    ) * 100, 2
  ) AS otif_like_percent,
  ROUND(AVG(late_days::numeric), 2) AS avg_late_days
FROM retail.order_analytics
WHERE purchase_date IS NOT NULL;
```

Figure A1: SQL-Based Fulfilment KPI Results (Benchmark Dataset)

This figure illustrates the monthly demand and on-time delivery performance derived from SQL, which is later used as input for dashboard development.

```
SELECT * FROM retail.monthly_demand
LIMIT 12;
```

Figure A2: Monthly Demand and Fulfilment Performance (SQL Output)

AI outputs were validated qualitatively against KPI patterns and operational logic; no automated decisions were executed without human approval

Summary:

Positioned AI as a decision-support capability to improve operational insight and exception management. AI outputs were validated against Excel KPIs and used to strengthen the root-cause narrative behind low OTIF performance. The outcome is an AI-enabled improvement backlog focused on risk scoring, supplier reliability monitoring, and automated exception analytics.

STEP 9: KPI & Fulfilment Analysis (Excel + SQL + Dashboard-ready)

9.1 Purpose of KPI and Dashboard Analysis:

The purpose of this step is to analyse order fulfilment performance and demand behaviour using both operational KPIs (derived from Excel-based analysis) and large-scale benchmarking insights (derived from SQL analysis). The analysis aims to identify the root causes behind service-level failures, particularly low OTIF performance, and to present these insights through interactive dashboards designed for decision-makers.

Power BI is used as the primary visualisation tool to convert analytics-ready datasets into intuitive, executive-facing dashboards that support monitoring, investigation, and scenario-based decision-making.

The detailed operational KPI model is developed in the analytical layer of the project, while Power BI is used to visualise aggregated trends, benchmark fulfilment performance, and dashboard-based decision support for stakeholders.

9.2.1 KPI Summary Interpretation

The Excel-based KPI summary indicates mixed operational performance:

Forecast Accuracy (MAPE = 3%)

Forecast accuracy is strong, indicating that demand prediction is not a major source of operational risk.

These operational KPI findings are derived from the project's simulated analytical model and should be interpreted separately from the external SQL benchmark metrics presented later in this section.

On-Time Delivery (79%)

A majority of orders are delivered on time; however, performance falls short of best-in-class benchmarks.

Stockout Rate (63%)

A high proportion of periods experience under-delivery relative to demand, signalling availability issues.

OTIF Performance (25%)

OTIF is significantly lower than both forecast accuracy and on-time delivery, indicating that service failures are driven by execution issues rather than demand planning.

OTIF Definition Used:

OTIF_Flag = 1 only when **OnTime_Flag = 1 AND Stockout_Flag = 0** (on-time and in-full).

9.2.2 Exception-Based Analysis

Exception periods were identified where either:

- Stockout_Flag = 1, or
- OnTime_Flag = 0

These exception months account for the majority of OTIF failures and highlight that even when deliveries are timely, under-delivery frequently prevents orders from being fulfilled in full.

This confirms that **supplier under-delivery and execution reliability**, rather than forecast error, are the dominant contributors to poor OTIF performance.

9.3 KPI Analysis – SQL Benchmark Dataset

9.3.1 SQL-Based Fulfilment KPIs

To provide external context, SQL was used to analyse a large-scale e-commerce dataset (Olist). The following benchmark KPIs were derived from the order_analytics view:

- **On-Time Delivery Percentage:** 90.45%
- **OTIF-like Performance:** 90.45%
- **Average Late Days:** 0.72 days

These results indicate that mature e-commerce operations can consistently achieve high delivery reliability with only marginal delivery delays.

9.3.2 Benchmark Interpretation

The SQL benchmark demonstrates that delivery timeliness can exceed 90% when operational execution is stable. Compared to the simulated operational dataset (On-Time 79%, OTIF 25%), the gap highlights that the primary improvement opportunity lies in execution reliability, particularly supplier fulfilment and delivery consistency.

The SQL dataset is used strictly as a **benchmarking and validation layer** and is not combined with the Excel KPIs.

9.4 Dashboard Development – Power BI

9.4.1 Dashboard Objectives

The Power BI dashboards were designed to:

- Provide executives with a high-level view of demand and fulfilment performance
- Enable operations teams to investigate service-level exceptions
- Support scenario analysis through interactive filters, targets, and bookmarks

9.4.2 Executive Overview Dashboard

The **Executive Overview** dashboard presents a consolidated view of performance using four headline KPIs:

- Total Orders
- Total Revenue
- Average Order Value
- On-Time Delivery Percentage

These measures are drawn from the aggregated monthly dashboard dataset and are intended to support executive visibility rather than replace the detailed analytical model.

Trend visuals show how order volumes, revenue, and on-time delivery evolve over time. A date slicer allows users to focus on specific periods and observe how service levels behave during demand peaks.

A Key Insights text panel summarises the most important observations, ensuring the dashboard supports storytelling rather than raw data inspection.

9.4.3 Operations & Exception Analysis

An operations-focused view enables deeper analysis of fulfilment behaviour:

- Trends in orders and revenue reveal demand seasonality
- On-time performance trends highlight periods of service degradation
- Scenario controls (Normal Demand vs Peak Demand) allow users to simulate operational stress conditions
- A what-if slider enables adjustment of target on-time performance to evaluate service gaps

These features support root-cause exploration and prioritisation of improvement actions.

9.4.4 Dashboard Acceptance Criteria:

- Cards must reconcile with the agreed KPI definitions and the underlying dashboard dataset.
- Slicer filters must affect all visuals consistently.
- Scenario buttons must change KPI view correctly.
- Insights box must update with scenario selection.

9.5 Interactive Features and Storytelling Elements

The dashboard incorporates several interactive elements to enhance decision support:

- **Slicers:** Enable time-based filtering
- **Bookmarks:** Allow switching between Normal Demand and Peak Demand scenarios
- **What-If Parameter:** Simulates changes in on-time performance targets
- **Dynamic Insight Text:** Updates contextually to guide interpretation

These features ensure the dashboard functions not only as a reporting tool but as an exploratory analytics interface.

9.6 Summary of Findings

The combined KPI and dashboard analysis leads to the following conclusions:

- Forecasting performance is strong and does not require replacement
- OTIF underperformance is primarily driven by execution failures rather than demand error
- Supplier under-delivery and delivery delays are the key operational risks
- Benchmark analysis confirms that significant improvement potential exists
- Interactive dashboards provide an effective mechanism for ongoing monitoring and exception management

9.7 Key Insights Summary

The analysis generated three core insights. First, forecast accuracy was relatively strong, indicating that demand prediction was not the primary performance constraint. Second, fulfilment outcomes remained weak, driven by frequent stockouts and inconsistent execution. Third, benchmark analysis showed that higher delivery reliability is achievable in practice, suggesting that the main improvement opportunity lies in operational execution rather than further forecast refinement.

STEP 10 — Recommendations, Implementation Plan & Change Management

10.1 Purpose of the Implementation Plan

The purpose of this step is to translate analytical insights into practical, implementable actions that improve order fulfilment performance. The recommendations focus on addressing the root causes of low OTIF performance identified in Step 9, while ensuring changes are realistic, scalable, and aligned with operational constraints.

10.2 Key Findings Driving Recommendations

The recommendations are based on the following validated insights:

- Forecast accuracy is strong (MAPE = 3%) and does not require replacement.
- OTIF performance is low (25%) despite relatively high on-time delivery (79%).
- Stockouts and supplier under-delivery are the dominant contributors to service failures.
- SQL benchmarking indicates that delivery timeliness above 90% is achievable in mature operations.
- Scenario analysis shows that fulfilment performance degrades under peak demand conditions.

These recommendations are informed by both the operational KPI model and the external fulfilment benchmark, ensuring that actions are grounded in internal diagnosis as well as broader performance context.

Recommendation 1 — Introduce Rule-Based Replenishment Controls

Objective: Reduce reliance on judgement-based planning and improve consistency in replenishment decisions.

- Define reorder thresholds using demand, lead time, and safety stock assumptions.
- Use weekly planning reviews to validate exceptions rather than every order manually.
- Standardise replenishment rules across the initial product scope.

Expected Impact: Improved availability and higher OTIF.

Expected Impact: Improved availability and higher OTIF.

Recommendation 2 — Implement Supplier Reliability Monitoring

Objective: Address execution risk at the supplier level.

- Track under-delivery rate (Delivered vs Ordered).
- Classify suppliers into risk tiers (Low / Medium / High).
- Prioritise corrective actions for high-risk suppliers.

Expected Impact: Reduced stockout frequency and more predictable fulfilment.

Recommendation 3 — Deploy Exception-Based Management

Objective: Focus operational attention where it matters most.

- Use dashboard thresholds (e.g., OTIF < target, Stockout_Flag = 1).
- Trigger review workflows only for exception periods.
- Reduce manual monitoring overhead.

Expected Impact: Faster issue resolution and better resource allocation.

Recommendation 4 — Use AI as Decision Support, Not as an Automated Control Mechanism

Objective: Enhance insight without replacing validated models.

- Use AI to highlight risk patterns and probable root causes.
- Validate AI outputs against KPI evidence before action.
- Avoid automated decision-making without human oversight.

Expected Impact: Better decisions without introducing operational risk.

10.4 Phased Implementation Roadmap:

Phase	Timeline	Key Activities	Deliverables
Phase 1: Foundation	Month 1	Finalise KPI definitions, Validate dashboards	KPI governance, baseline dashboard
Phase 2: Control	Month 2	Implement replenishment rules, supplier metrics	Rule-based planning, supplier scorecards
Phase 3: Intelligence	Month 3	Deploy AI-assisted exception analysis	Risk insights, exception reports
Phase 4: Optimisation	Month 4	Review Performance, refine thresholds	Improved OTIF, continuous improvement plan

10.5 Change Management Considerations

Successful implementation requires stakeholder buy-in and controlled change.

Key Change Actions:

- Train planners and operations teams on KPI interpretation.
- Communicate clearly that AI supports decisions, not replaces judgement.
- Align incentives with OTIF improvement rather than order volume alone.

Risks & Mitigation:

Risks	Mitigation
Resistance to new rules	Pilot rules in one category before scaling
Over-reliance on AI	Enforce KPI Validation gates
Data Quality degradation	Regular data audits and exception checks

10.6 Success Metrics & Monitoring

Post-implementation success will be monitored using:

- OTIF improvement trend
- Reduction in Stockout Rate
- Supplier under-delivery reduction
- Stability of On-Time Delivery during peak demand

Targets should be reviewed quarterly and adjusted based on operational learning.

10.7 Alignment with Business Analysis & Consulting Practice

This implementation plan reflects standard consulting practice by:

- Linking analysis directly to business decisions
- Prioritising high-impact, low-risk interventions
- Phasing changes to manage complexity
- Embedding analytics into operational workflows

10.8 Summary

The recommendations focus on improving execution reliability rather than replacing strong forecasting capability. By combining rule-based controls, supplier monitoring, exception-

driven management, and AI-assisted insights, the organisation can realistically improve OTIF performance while maintaining operational stability.

STEP 11 — Review, Retrospective & Continuous Improvement

11.1 Purpose of the Review and Retrospective

The purpose of this step is to evaluate the effectiveness of the implemented analytical solution and identify opportunities for continuous improvement. Rather than treating the project as a one-off delivery, this step adopts an Agile and consulting-oriented mindset, focusing on learning, refinement, and sustained performance improvement.

11.2 Performance Review Against Objectives

The outcomes of the project were reviewed against the original business objectives defined in Step 1.

Objective	Outcome
Improve forecast accuracy	Achieved (MAPE = 3%)
Improve fulfilment visibility	Achieved through KPI framework and dashboards
Identify root causes of low OTIF	Achieved (stockouts and execution issues identified)
Enable decision support	Achieved via Power BI scenarios and AI insights

The review confirms that the analytical objectives were met and that the solution provides actionable insights rather than descriptive reporting alone.

The key learning from this engagement is that improving execution reliability delivers materially higher returns than further forecast refinement once baseline accuracy is achieved.

11.3 Sprint Retrospective (Agile Perspective)

A retrospective was conducted to evaluate the delivery process and identify improvements for future iterations.

Objective	Evidence	Status
Forecast accuracy	MAPE 3%	Achieved
Fulfilment visibility	Dashboards	Achieved
Root Cause clarity	Stockout + delays	Achieved

What Went Well

- Clear problem definition and KPI alignment
- Strong separation of forecasting accuracy and execution reliability

- Effective use of dashboards for stakeholder communication
- Successful integration of AI as a decision-support tool

What Could Be Improved

- Earlier validation of supplier delivery data
- More granular lead-time analysis
- Earlier stakeholder feedback on dashboard usability

11.4 Actions for Next Iteration

- Incorporate supplier-level data into the analytics layer
- Add automated alerts for repeated exception patterns
- Enhance dashboard usability based on user feedback

Area	Improvement Opportunity
Data	Introduce SKU-level and supplier-level granularity
Analytics	Add predictive risk scoring for OTIF failures
Automation	Automate exception alerts and reporting
Decision Support	Integrate what-if simulations for inventory buffers

These initiatives allow the solution to evolve from descriptive analytics towards predictive and prescriptive capability.

11.5 Governance and Ownership

To ensure sustainability, ownership of key components is defined:

- **Business Owner:** Operations Manager
- **Data Owner:** Analytics / BI Team
- **Decision Authority:** Supply Chain Leadership

Clear ownership ensures accountability and consistent use of insights in operational decision-making

11.6 Alignment with Professional Practice

This review process aligns with professional Business Analysis and Consulting standards by:

- Treating analytics solutions as living systems
- Embedding learning loops through retrospectives
- Linking insight generation with operational ownership
- Supporting data-driven continuous improvement

11.7 Summary

The review and retrospective confirm that the project delivered meaningful business insight and decision-support capability. By adopting an iterative mindset and defining a clear improvement roadmap, the solution remains adaptable to changing demand patterns and operational challenges.

11.8 Decision Log:

- Decision 1: Keep AI as decision support (not automation).
- Decision 2: Focus improvements on supplier under-delivery.
- Decision 3: Maintain weekly exception review cadence.

STEP 12 — Roadmap & Future Enhancements

12.1 Purpose of Continuous Improvement

This step ensures that the analytical solution is not treated as a one-off exercise but as a continuously evolving capability. The objective is to embed learning, refinement, and scalability into the order fulfilment and demand forecasting process, aligning with Agile delivery and UK corporate best practices.

12.2 Lessons Learned

- Forecast accuracy alone does not ensure OTIF success
- Execution reliability is the key constraint (supplier under-delivery + late delivery)
- Dashboards are most valuable when exception-driven
- AI adds value as a prioritisation and explanation tool

12.3 Data & Analytics Maturity Path

The solution can evolve across analytics maturity levels:

Stage	Description
Descriptive	KPI dashboards and historical trends
Diagnostic	Root-cause analysis of OTIF failures
Predictive	AI-assisted risk scoring and demand anomalies
Prescriptive	What-if simulations and decision recommendations

This progression aligns with enterprise analytics roadmaps used in UK organisations.

12.4 Governance for Continuous Improvement

To ensure sustainability:

- KPI definitions and thresholds should be reviewed quarterly.
- Dashboard changes should follow a controlled change process.
- AI models should be periodically validated against actual outcomes.
- Data quality checks should remain part of Sprint 0 for future iterations.

Clear governance prevents metric drift and maintains stakeholder trust.

12.5 Alignment with Business Analysis & Consulting Practice

This continuous improvement approach reflects professional consulting standards by:

- Maintaining a prioritised backlog
- Delivering incremental value through sprints
- Linking analytics directly to business decisions
- Treating insights as inputs to action, not endpoints

12.6 Summary

The continuous improvement roadmap positions the solution as a scalable analytics capability rather than a static report. By extending granularity, automating exception management, and enhancing AI-assisted insights, the organisation can progressively improve fulfilment reliability and decision quality.

STEP 13 — Final Portfolio Packaging & Professional Presentation

13.1 Purpose of Final Packaging

The purpose of this step is to present the project outputs in a professional, employer-ready format suitable for academic submission, job applications, interviews, and consulting discussions. The focus is on clarity, credibility, and alignment with UK corporate expectations for Business Analysts and Consultants.

13.2 Final Deliverables

The completed project includes the following artefacts:

- **Consulting-style report (PDF)**
End-to-end documentation covering problem definition, analysis, insights, and recommendations.

- **Analytical Modelling Layer**
Forecasting logic, KPI calculations, data validation rules, and supporting assumptions used to derive operational insight.
- **SQL Analytics Layer (PostgreSQL)**
Benchmark fulfilment analysis using real e-commerce data to validate operational patterns.
- **Power BI Interactive Dashboard**
Executive Overview and Operations & Exceptions views with scenario-based storytelling.
- **AI Use-Case Documentation**
AI-assisted exception detection with explicit validation against KPI evidence.

These artefacts together demonstrate a full analytics lifecycle.

13.3 Portfolio & Employability Positioning (UK Context)

This project can be positioned in the UK job market as:

- A Business Analyst portfolio project showing requirements, KPIs, and stakeholder thinking
- A Consulting-style case study with insight-to-action translation
- Evidence of Excel, SQL, Power BI, and AI literacy
- Demonstration of Agile delivery and continuous improvement

13.4 How to Present This in Interviews (Talking Points)

When discussing this project, the following structure can be used:

1. **Business Problem**
“Forecast accuracy was strong, but OTIF performance remained low due to execution issues.”
2. **Analytical Approach**
“I combined Excel forecasting, KPI design, SQL benchmarking, and Power BI dashboards.”
3. **AI Positioning**
“AI was used for exception detection and decision support, not automation.”
4. **Outcome**
“The analysis identified stockouts and supplier under-delivery as primary OTIF drivers.”
5. **Impact**
“Recommendations focused on rule-based replenishment, supplier monitoring, and exception management.”

This narrative aligns closely with UK corporate interview expectations.

13.5 Evidence & Appendices Structure

The project evidence is organised as follows:

Appendix A — SQL Evidence

- Fulfilment KPI query results
- Monthly demand aggregation output

Appendix B — Power BI Dashboards

- Executive Overview dashboard
- Operations & Exceptions dashboard
- Scenario filter example

Appendix C — Analytical Model Evidence

- Simulated operational dataset
- KPI logic and summary outputs
- Forecast model and accuracy metrics

This structure ensures transparency and traceability.

13.6 Reflection on Professional Readiness

This project demonstrates readiness for a Business Analyst or Consulting role by showing:

- Structured problem-solving
- Data-driven decision making
- Stakeholder-focused communication
- Responsible and explainable use of AI
- Ability to convert insights into actionable recommendations

13.7 Summary

The final project is positioned as both an academic submission and a professional portfolio asset. It demonstrates an end-to-end business analysis lifecycle, integrates modern analytics tools, and reflects UK corporate and consulting best practices.